

## SOA Survey

## Early Warning System for Forest Fire

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**Abstract**

Forest fires pose one of the most destructive natural hazards globally, threatening ecosystems, wildlife, and human settlements every year. Early detection before a fire grows uncontrollable is critical, yet traditional approaches relying on watchtowers, satellite imagery, and manual inspection suffer from fundamental limitations in spatial coverage, detection latency, and operational cost. This survey reviews the state of the art in deep learning-based forest fire and smoke detection, synthesizing fifteen papers spanning image-based detection, video analysis, lightweight model design, dataset construction, and vision-language integration. We examine core architectures including YOLO variants, Faster R-CNN, attention-based CNNs, and hybrid transformer models. Our review identifies three central gaps: most high-accuracy models are too computationally demanding for real-time deployment on edge hardware such as the Raspberry Pi or Jetson Nano; the persistent confusion between early-stage smoke and environmental visual confounders such as clouds, mist, and steam remains unsolved in fixed-camera deployments; and nighttime and low-light fire detection is severely underrepresented in both datasets and published methods. We propose investigating a lightweight detection model augmented with CLIP-based features and trained with synthetic nighttime augmentation, evaluated on a custom dataset annotated from WWF-deployed PTZ cameras.

**1. Problem Definition**

Forest fires are among the most severe and rapidly escalating natural disasters worldwide. Hundreds of millions of hectares of forests are devastated annually, causing irreversible ecological damage, significant loss of human life, and enormous economic costs [1]. Studies indicate that approximately 85% of wildfires in the United States between 1992 and 2015 were caused by human activity, underscoring that effective early-warning infrastructure can prevent the majority of destructive fires [2]. Once a fire ignites, the window for intervention narrows drastically: a small ignition

on a high-risk day can grow out of control within minutes, making early detection of the first signs of smoke the single most impactful intervention point available.

This project addresses the problem of automated fire and smoke detection using footage from Pan-Tilt-Zoom (PTZ) cameras deployed at fixed vantage points across forest sites. The task can be formulated in one of three ways depending on the desired level of output: (1) *binary classification*, where each image or video frame is labelled as fire/smoke present or absent; (2) *bounding box detection*, where the spatial location of fire or smoke within the frame is also predicted; or (3) *instance segmentation*, where a pixel-level mask is produced. The system must operate in real time and be deployable on edge hardware such as a Raspberry Pi or NVIDIA Jetson device, since the cameras are located at remote forest sites with limited connectivity.

What makes this problem uniquely challenging from a deep learning perspective is a combination of factors. First, early-stage smoke is visually ambiguous — it occupies a small fraction of image pixels, has poorly defined edges, and its appearance (transparency, colour, texture) overlaps with benign phenomena such as clouds, haze, and dust [7]. Second, PTZ cameras are mounted at fixed elevated positions looking across the horizon, meaning that clouds, morning mist, and distant smoke appear in the same region of the frame at the same altitude, making appearance-only classification unreliable [10]. Third, the computational budget is severely constrained: a model that achieves excellent accuracy on a high-end GPU may be unusable on an edge device operating at 5–10 frames per second. Fourth, the problem extends beyond daylight hours — fires ignited at night or smouldering into the night are effectively invisible to standard RGB models trained exclusively on day-

time data, and nighttime datasets remain scarce [2]. Finally, annotated datasets of early-stage forest fire footage from fixed cameras in diverse geographic regions remain limited, making generalisation to new deployment sites such as South Asian forest terrain difficult [12].

## 2. Literature Review

This review organises the surveyed literature around the core challenges facing the field rather than cataloguing papers individually. Six challenges are identified: detection architecture design, early-stage smoke and false-alarm suppression, cloud and smoke confusion, nighttime and low-light detection, edge deployment and model efficiency, and dataset availability and standardisation. The emerging role of vision-language models is discussed last as a cross-cutting theme.

### 2.1. Detection Architectures: From Sensor Fusion to Deep CNNs

Early work on automated fire perception relied on multi-sensor fusion rather than data-driven learning. Martinez-de Dios et al. [1] presented a system that combined visual and infrared (IR) cameras with GPS telemetry to produce a real-time 3D model of a fire front, measuring flame height, inclination angle, and base width simultaneously. Fuzzy-wavelet filters were applied to suppress noise in the IR signal while preserving the progressive spread of the fire front. Validated on field experiments conducted in Gestosa, Portugal, between 1998 and 2005, this system demonstrated that complementary information from visual and infrared sensors substantially improves robustness over single-modality approaches. However, the method is heavily hand-engineered, requiring manual calibration and explicit feature design, and does not generalise without significant re-engineering to new sites.

A follow-on line of work retained the IR modality but replaced hand-crafted features with algorithmic segmentation. The approach in [3] processes infrared UAV imagery by applying histogram-based segmentation to extract hot candidate regions, then uses optical flow to compute motion vectors for each candidate. Fire regions are distinguished from fire analogues

through morphological operations and a blob counter. This pipeline achieves real-time fire tracking within individual IR frames but requires a calibrated IR sensor and is sensitive to abrupt camera movement.

The field shifted decisively toward deep learning following the success of convolutional neural networks. The comprehensive review by Bouguettaya et al. [2] surveyed approximately 40 papers published between 2013 and 2021 and documented the transition from hand-crafted RGB colour thresholds to end-to-end CNN architectures. The dominant paradigm moved to one-stage detectors: multiple versions of YOLO, SSD, and two-stage Faster R-CNN, as well as segmentation networks such as U-Net and temporal models including LSTM and deep belief networks. Subsequent work refined these architectures for the fire domain. Kim and Muminov [5] enhanced YOLOv7 with CBAM attention, SPPF+, BiFPN, and decoupled heads, achieving AP50 of 86.4% on UAV smoke images. The evaluation in [6] further reveals that YOLOv8 achieves recall of 0.9 for fire but only 0.7 for smoke, quantifying the difficulty of early smoke detection relative to visible flame.

### 2.2. Early-Stage Smoke: Small Targets and False Alarms

Detecting fire at the earliest possible stage means detecting smoke, often before a visible flame exists. The Wildfire Smoke Detection System in [7] introduces the SKLFS-WildFire dataset, which specifically captures fires in their initial stage with over 299,827 training images and proposes Cross Contrast Patch Embedding (CCPE) to improve low-level smoke feature extraction in transformer architectures. The authors demonstrate that standard transformers are insufficient for this task because early-stage smoke texture and transparency are encoded in low-level feature maps that transformer patch embedding tends to discard. The Separable Negative Sampling Mechanism (SNSM) additionally reduces false positives from background clutter.

A continual learning variant of YOLOv3 [8] incorporates a Feature Squeeze Block under a learning-without-forgetting framework, achieving approximately 98% detection precision while retaining knowledge of previously seen deployment sites. This

technique is particularly relevant when a system must be updated with new site-specific data without full re-training, as will be necessary when in the case of new WWF camera sites or deployment in a new location. The comprehensive review in [9] synthesises findings from 80 studies and consistently identifies high false alarm rates as the primary unsolved challenge across the literature.

### 2.3. Cloud and Smoke Confusion in Fixed-Camera Deployments

A particularly severe failure mode in fixed-camera setups is the visual confusion between early-stage smoke and environmental phenomena such as clouds, morning mist, steam from water bodies, and haze. Unlike UAV systems that often look downward toward a ground reference, PTZ cameras mounted at elevated forest positions look across the horizon, placing clouds and distant smoke in the same spatial region of the frame at the same apparent altitude. This makes appearance-only classification unreliable at the most critical early-detection window — dawn and dusk, when mist and low cloud are most prevalent.

The benchmarking study in [10] explicitly addresses this problem by constructing a multi-scene benchmark with hard negative annotations for clouds, sunset conditions, morning mist, and steam. The study demonstrates that models trained without these explicit hard negatives produce false alarm rates 30–40% higher than models trained with them, confirming that the standard practice of simply omitting confounders from training data is insufficient. The analysis in [14] further shows that compact models (YOLOv5n, YOLOv8n) are particularly susceptible to this failure mode due to their limited representational capacity.

The cloud/smoke confusion problem operates at three distinct levels, each requiring a different mitigation strategy. At the colour and texture level, both clouds and smoke are grey-white with soft undefined edges; explicit hard-negative training is the primary defence. At the motion level, clouds drift uniformly across the frame while smoke rises from a fixed ground point source and disperses irregularly; temporal models that analyse consecutive frames can exploit this motion difference [11]. At the contextual level, smoke origi-

nates from a single location on the ground and its trajectory follows wind direction, whereas clouds maintain consistent altitude over longer time windows; scene-level context modelling addresses this layer but remains largely unexplored. Critically, this contextual understanding is precisely where vision-language models such as CLIP offer a concrete advantage: by encoding descriptions such as “smoke rising from burning trees” versus “clouds drifting over a mountain,” CLIP provides a semantic disambiguation layer that pure visual detectors fundamentally lack [15].

### 2.4. Nighttime and Low-Light Fire Detection

Nighttime fire detection is one of the most under-represented problems in the field. The majority of published datasets and models assume daylight conditions; Bouguettaya et al. [2] note this as an explicit gap, and the 20-year dataset review in [12] confirms that night-time and low-light annotated footage constitutes a small minority of all available data. This is a practical problem: fires that ignite or smoulder at night are invisible to RGB models trained exclusively on daytime imagery, and the absence of colour cues means that the orange/red flame signatures that dominate appearance-based detectors disappear entirely.

FLAME 2 is currently the most relevant publicly available dataset for this sub-problem, providing seven paired RGB and infrared (IR) video sequences captured during wildfire events. At night, IR sensors detect heat signatures rather than visible light, making them substantially more reliable than RGB cameras for flame localisation. However, early-stage *smoke* at night does not yet produce a detectable heat signature, meaning IR alone is insufficient for the earliest warning window.

A practical approach for projects with limited access to nighttime data is synthetic augmentation: applying heavy darkening, blue moonlight tinting, Gaussian sensor noise, and fire-region-preserving darkness filters to existing daytime footage to create simulated nighttime training samples. When combined with CLAHE contrast enhancement (which approximates the behaviour of IR cameras by amplifying local contrast differences), this approach can substantially increase nighttime robustness without requiring a dedicated nighttime collection campaign [13]. The key

training strategy is staged fine-tuning: a model first trained on large daytime datasets, then fine-tuned on available nighttime data (FLAME 2 IR sequences), and finally adapted on site-specific footage with synthetic nighttime augmentation applied to daytime WWF frames.

## 2.5. Edge Deployment and Lightweight Model Design

A critical constraint for this project is that inference must run in real time on an edge device co-located with the PTZ camera. Jadon et al. [4] were among the first to directly target this constraint by designing FireNet, a custom CNN built from scratch to balance detection accuracy against model size for Raspberry Pi deployment. More recent work by [13] applies knowledge distillation to transfer accuracy from a powerful teacher model to a compact MobileViT-S student, achieving strong detection performance while meeting the computational budget of UAV-mounted edge processors with concrete FPS benchmarks on embedded hardware.

An important practical note for this project is that edge device performance benchmarks reported in the literature were obtained on physical Jetson Nano or Raspberry Pi hardware. When direct access to such devices is unavailable during development, as in the case of this project, laptop-based simulation under constrained conditions (CPU thread capping, limited VRAM allocation) provides a directional estimate but must not be presented as equivalent to real hardware performance. Any reported inference speed should explicitly state whether it was measured on target hardware or simulated.

## 2.6. Datasets and Benchmark Standardisation

The quality and availability of annotated datasets is a bottleneck across all sub-problems. The most extensive audit is provided by [12], which reviews datasets spanning 20 years. FIGLib, introduced by Dewangan et al. [11], is particularly relevant to this project as it was collected from fixed-view cameras analogous to PTZ setups. The dataset contains 315 fire sequences from 101 cameras in Southern California with 24,800 high-resolution images. The SmokeyNet architecture exploits temporal structure through spatiotempo-

ral feature aggregation, confirming that consecutive-frame analysis substantially outperforms single-frame detection for early smoke.

The benchmarking study in [10] identifies inconsistent annotation standards across public datasets as a primary source of unreliable cross-paper comparisons. FASDD provides over 120,000 heterogeneous images and is particularly useful for generalisation testing because of its geographic and environmental diversity. A notable limitation across all public datasets is geographic bias: most are collected in North America or Europe and do not reflect the vegetation, terrain, and atmospheric conditions of South Asian forest environments targeted by this project, reinforcing the necessity of a custom WWF dataset.

## 2.7. Vision-Language Models for Contextual Fire Detection

The most recent direction in the literature is the integration of large vision-language models (VLMs) to address both the generalisation and the false-alarm problems simultaneously. Challa et al. [15] apply CLIP and LLaVA in a zero-shot learning framework for forest fire detection, establishing that pre-trained VLMs can classify fire/smoke images without any domain-specific fine-tuning by leveraging natural language descriptions. While zero-shot performance falls 10–20% below fine-tuned supervised methods, the paper demonstrates that combining CLIP features with a fine-tuned detection head outperforms both approaches individually.

The most practically useful role of CLIP in this project’s pipeline is as a contextual disambiguation layer for low-confidence detections. When a compact YOLO model produces a detection with confidence in the 0.3–0.6 range — the ambiguous zone where cloud/smoke confusion is most likely — the detection crop can be passed to CLIP with targeted text prompts: “forest smoke rising from burning trees,” “cloud drifting over mountain,” “morning mist in a valley.” CLIP’s vote, which encodes semantic and contextual understanding beyond pixel-level appearance, then adjusts the final detection confidence. This is especially powerful at dawn and dusk, and for nighttime glow disambiguation (“fire glowing in dark forest” versus “vehicle headlight through trees”), without

requiring nighttime training data for CLIP itself.

### Open Problems (flagged by the field):

- False alarm rates from visual confounders (clouds, mist, sunset) remain the dominant unsolved challenge in deployed fixed-camera systems; no published method has achieved reliable performance under these conditions on held-out deployment sites [10, 9].
- Cloud and smoke confusion in horizon-facing fixed cameras is a distinct and largely unaddressed sub-problem; most datasets do not include sufficient hard negative examples of these confounders [10].
- Nighttime and low-light fire detection lacks both adequate datasets and validated architectures; the field has identified it as a gap but no published solution has been demonstrated on fixed-camera forest footage [2, 12].
- Domain shift from public datasets to new geographic deployments (e.g. South Asian forest terrain) is not solved by pre-processing alone and requires localised fine-tuning on site-specific footage, which is rarely studied [11].
- Standardised evaluation benchmarks combining multiple datasets, consistent annotation formats, and realistic positive-to-negative ratios do not yet exist [12].
- Efficient integration of temporal information from video on edge hardware has not been demonstrated at acceptable latency [11].

### 3. Research Gap and Proposed Direction

The literature review exposes three compounding gaps. First, on the accuracy side, high-performance models such as enhanced YOLOv7 [5] and transformer-based detectors [7] require GPU-class hardware and have not been validated on fixed PTZ camera footage in the geographic and environmental conditions of this project. Second, on the deployment side, lightweight models [4, 13] sacrifice accuracy and are particularly unreliable for the harder sub-tasks of early-stage smoke and visually ambiguous cloud/smoke discrimination. Third, on

the temporal and lighting side, nighttime and low-light detection remains an open sub-problem with no published solution validated on fixed-camera forest footage, and the interaction between domain shift, staged fine-tuning, and synthetic nighttime augmentation has not been studied in a single unified pipeline.

The WWF PTZ camera footage itself constitutes an additional gap: no public dataset covers this geographic region, camera hardware, or forest type. FigLib is the closest analogue but is limited to Southern California. Dataset construction and multi-condition annotation (daytime, dawn/dusk, nighttime) from the raw PTZ footage is therefore a necessary contribution in its own right.

Prior work achieves strong detection on curated benchmarks but struggles with early-stage smoke, cloud/smoke confusion, and nighttime conditions on edge hardware in real forest deployments; we therefore propose to investigate a lightweight fire and smoke detection model augmented with CLIP-based contextual features, trained with staged fine-tuning and synthetic nighttime augmentation, and evaluated on a custom annotated dataset from WWF-deployed PTZ cameras under both daytime and low-light conditions.

**Research Question (formal):** Can a lightweight deep learning model, augmented with CLIP-based contextual features and trained via staged fine-tuning with synthetic nighttime augmentation, achieve real-time accurate early-stage fire and smoke detection from fixed PTZ camera footage simulated on edge hardware, while outperforming standard compact-model baselines on a custom WWF forest fire dataset?

### 4. Benchmark, Evaluation, and Data Landscape

**Datasets.** Table 1 summarises the key publicly available datasets identified in the survey. FigLib [11] and SKLFS-WildFire [7] are the most directly relevant as primary training supplements. D-Fire serves as the standard compact-model comparison benchmark [14]. FLAME 2 is the primary dataset for nighttime/IR training and FASDD provides the largest heterogeneous collection for generalisation testing.

Table 1: Key publicly available fire and smoke datasets.

| Dataset     | Size      | Modality | Notes                                                        |
|-------------|-----------|----------|--------------------------------------------------------------|
| D-Fire      | 21K+ img  | RGB      | Mixed outdoor; compact-model benchmark                       |
| FIgLib      | 24.8K img | RGB      | Fixed-view cameras; closest to PTZ setup                     |
| SKLFS       | 350K+ img | RGB      | Real early-stage wildfire footage                            |
| FLAME 2     | 7 videos  | RGB + IR | Paired visual-IR; primary nighttime/low-light dataset        |
| FASDD       | 120K+ img | RGB      | Heterogeneous; geographic diversity; good for generalisation |
| VisiFire    | 40 clips  | RGB      | Forest and urban scenes                                      |
| FIRESENSE49 | 49 videos | RGB      | Heritage and urban environments                              |

**Evaluation Protocol.** The field uses mAP at IoU 0.5 (mAP<sub>50</sub>) as the primary detection metric, with Precision, Recall, and F1-score alongside it. For this project, Frames Per Second (FPS) on the target edge device is an equally critical metric. False Positive Rate (FPR) under confounder conditions (clouds, sunset, mist) must be reported separately, as overall mAP masks the false-alarm problem that is the dominant failure mode in deployment [10]. Additionally, performance under simulated nighttime conditions should be reported as a distinct evaluation partition.

**Compute Context.** Published methods primarily train on NVIDIA V100 or A100 GPUs. Inference benchmarks in the edge deployment literature are conducted on NVIDIA Jetson Nano (472 GFLOPS) and Raspberry Pi 4 ( $\approx 13.5$  GFLOPS) [13, 4]. Quantised and distilled models typically achieve 10–30 FPS on Jetson Nano; unoptimised YOLOv8 variants achieve approximately 3–8 FPS on the same hardware.

## Experimental Design Considerations

- **Baselines.** Any proposed method must outperform at minimum YOLOv8n [14] and FireNet [4] on both mAP<sub>50</sub> and FPS on target edge hardware. A CLIP zero-shot baseline [15] must also be evaluated to quantify the benefit of fine-tuning over zero-shot inference. A daytime-only model trained without nighttime augmentation should be evaluated as a separate ablation point to isolate the contribution of nighttime training.
- **Ablation.** Key variables to isolate: (1) backbone size vs. accuracy trade-off; (2) marginal contribution of CLIP features over a purely visual backbone; (3) effect of including cloud/mist/sunset hard negatives on false alarm rate; (4) contribution of synthetic nighttime augmentation on low-light detection performance.
- **Evaluation pitfalls.** *Geographic leakage:* training and testing on the same camera site inflates generalisation estimates; at least one WWF camera site must be held out entirely [11]. *Positive-negative imbalance:* the 50/50 split in FIgLib does not reflect real deployment ratios; precision at low false alarm rates is more informative than mAP alone. *Confounder blindness:* clouds, sunset, and morning mist must be tested as an explicit evaluation partition; models routinely fail on these even when overall mAP is acceptable [10]. *Simulation vs. real hardware:* if Jetson or Raspberry Pi devices are unavailable, inference speed must be measured under CPU-constrained conditions and explicitly reported as simulated; GPU benchmark numbers must not be presented as edge-device performance.

## 5. Annotated Reading List

### References

- [1] J. R. Martinez-de Dios, B. C. Arrue, A. Ollero, L. Merino, and F. Gómez-Rodríguez. Computer vision techniques for forest fire perception. *Image and Vision Computing*, 26(5):550–562, 2008. *What it does: Fuses visual camera, IR camera, and GPS to build a real-time 3D fire model using fuzzy-wavelet filters. What it contributes: Demonstrates IR + RGB sensor fusion*

- improves measurement accuracy over single-modality approaches. Why it is here: Foundational sensor-fusion baseline; motivates multi-modal approaches and IR use for nighttime detection.*
- [2] A. Bouguettaya, H. Zarzour, A. M. Taberkit, and A. Kechida. A review on early wildfire detection from UAVs using deep learning-based computer vision algorithms. *Signal Processing*, 190:108309, 2022. *What it does: Surveys ~40 papers from 2013–2021 covering YOLO, R-CNN, SSD, U-Net, LSTM, and communication technologies. What it contributes: Most comprehensive landscape overview; identifies nighttime detection and edge complexity as key gaps. Why it is here: Primary landscape paper covering all architectures we compare against.*
- [3] (Authors confirmed via IEEE Xplore arXiv no. 7991306.) Fire detection using infrared images for UAV-based forest fire surveillance. *IEEE Conference Proceedings*, 2017. *What it does: Detects fire in UAV IR imagery via histogram segmentation and optical flow motion analysis. What it contributes: Shows brightness and motion in IR provides reliable fire localisation without deep learning. Why it is here: Classical IR baseline; motivates IR as a complement to RGB for nighttime fire localisation.*
- [4] A. Jadon, M. Omama, A. Varshney, M. S. Ansari, and R. Sharma. FireNet: A specialized lightweight fire & smoke detection model for real-time IoT applications. *arXiv:1905.11922*, 2019. *What it does: Designs a shallow CNN (FireNet) from scratch for Raspberry Pi real-time fire/smoke detection. What it contributes: Establishes the lightweight CNN edge baseline; open-sources model and dataset. Why it is here: Primary compact-model baseline for Raspberry Pi / Jetson deployment comparison.*
- [5] S.-Y. Kim and A. Muminov. Forest fire smoke detection based on deep learning approaches and unmanned aerial vehicle images. *Sensors*, 23(12), 2023. *What it does: Enhances YOLOv7 with CBAM, SPPF+, BiFPN, and decoupled heads on 6,500 UAV smoke images. What it contributes: AP50 of 86.4%; best published UAV smoke detection result in this survey. Why it is here: Upper-bound accuracy target before edge optimisation.*
- [6] (Authors from IJACSA Vol. 15 No. 1.) A YOLO-based approach for fire and smoke detection. *IJACSA*, 15(1), 2024. *What it does: Applies YOLOv8 and separately analyses precision-recall curves for fire vs. smoke classes. What it contributes: Quantifies the fire/smoke performance gap (recall 0.9 vs. 0.7 at same threshold). Why it is here: Provides YOLOv8 baseline and motivates special treatment for the smoke sub-task.*
- [7] (Authors from arXiv:2311.10116.) Wild-fire smoke detection system: Model architecture, training mechanism, and dataset. *arXiv:2311.10116*, 2023. *What it does: Introduces SKLFS-WildFire (350K+ early wildfire images) and CCPE + SNSM modules on YOLOX. What it contributes: Largest real early-stage wildfire dataset; shows transformers alone are insufficient for early smoke. Why it is here: Hardest sub-problem in our project; SKLFS supplements WWF data during training.*
- [8] (Authors from Fire Ecology, Springer Open.) Forest fire and smoke detection using deep learning-based learning without forgetting. *Fire Ecology*, 2022/2023. *What it does: YOLOv3 + Feature Squeeze Block under continual learning; ~98% precision retaining prior site knowledge. What it contributes: Demonstrates LwF prevents catastrophic forgetting when fine-tuning on new site data. Why it is here: Relevant when the model must be updated as new WWF camera sites come online.*
- [9] (Authors from MDPI Applied Sciences, 2025.) Early fire and smoke detection using deep learning: A comprehensive review. *Applied Sciences*, 15(18):10255, 2025. *What it does: Reviews 80 studies across CNNs, YOLO, Faster R-CNN, and spatiotemporal models. What it contributes: Identifies high false alarm rates as the dominant unsolved problem across the field. Why it is here: Cross-validates open problems identified in this survey; ensures complete coverage.*

- [10] (Authors from arXiv:2410.16631.) Benchmarking multi-scene fire and smoke detection. *arXiv:2410.16631*, 2024. *What it does: Unifies FIRESENSE, BoWFire, D-Fire with consistent labels and explicit confounder annotations. What it contributes: Shows confounder training reduces false alarms 30–40%; exposes annotation inconsistency across datasets. Why it is here: Directly informs our cloud/smoke hard negative strategy and evaluation protocol.*
- [11] A. Dewangan, Y. Pande, H.-W. Braun, F. Veron, I. Perez, I. Altintas, G. W. Cottrell, and M. H. Nguyen. FigLib & SmokeyNet: Dataset and deep learning model for real-time wildland fire smoke detection. *Remote Sensing*, 14(4):1007, 2022. *What it does: Introduces FigLib (24,800 images from 101 fixed-view cameras) and SmokeyNet spatiotemporal architecture. What it contributes: Closest PTZ-camera analogue; temporal model rivals human performance in smoke detection. Why it is here: Primary pre-training dataset; SmokeyNet is key architectural reference for temporal detection.*
- [12] (Authors from arXiv:2503.14552.) Fire and smoke datasets in 20 years: An in-depth review. *arXiv:2503.14552*, 2025. *What it does: Reviews every major fire/smoke dataset from 20 years across RGB, thermal, IR; benchmarks on ResNet-50, DeepLabV3, YOLOv8. What it contributes: Exposes cross-dataset performance variation; most comprehensive dataset audit available. Why it is here: Essential for dataset selection and understanding generalisation limits to new geographies.*
- [13] (Authors from arXiv:2502.20979.) Real-time aerial fire detection on resource-constrained devices using knowledge distillation. *arXiv:2502.20979*, 2025. *What it does: Compresses MobileViT-S via knowledge distillation for UAV edge processors with hardware FPS benchmarks. What it contributes: State-of-the-art edge compression; concrete latency numbers on embedded hardware. Why it is here: Edge compression technique to apply or compare against for Jetson / Raspberry Pi deployment.*
- [14] A. S. Joshi et al. Uncertainty-aware post-detection framework for enhanced fire and smoke detection in compact deep learning models. *arXiv:2510.10108*, 2025. *What it does: Evaluates YOLOv5n/YOLOv8n on D-Fire; proposes Confidence Refinement Network using uncertainty + colour/edge. What it contributes: Lightweight post-detection module improving compact-model performance without modifying the base model. Why it is here: Actionable technique for improving compact baseline; benchmarks on D-Fire our standard eval dataset.*
- [15] N. R. Challa, B. C. Mohan, and M. L. Babu. Enhancing forest fire detection using zero-shot learning: A LLaVA and CLIP vision-language model approach. *Multimedia Tools and Applications*, Springer, 2026. *What it does: Applies CLIP and LLaVA zero-shot for forest fire detection using natural language prompts. What it contributes: First zero-shot VLM baseline for forest fire; combining CLIP with fine-tuned head outperforms both alone. Why it is here: Motivates and benchmarks CLIP in our pipeline; defines zero-shot lower bound and cloud/smoke disambiguation use case.*

## 6. Preliminary Implementation Resources

- <https://github.com/ultralytics/ultralytics> YOLOv8/v11 repo. Primary backbone candidate with TensorRT export for Jetson deployment.
- <https://github.com/ultralytics/yolov5> YOLOv5 nano/small variants. Training infrastructure for compact-model baselines.
- <http://hpwren.ucsd.edu/HPWREN-FigLib/> FigLib dataset download. Primary fixed-camera pre-training dataset; closest public analogue to PTZ setup.
- <https://github.com/openai/CLIP> OpenAI CLIP (ViT-B/32). Visual-language embeddings for contextual cloud/smoke disambiguation and nighttime glow classification.
- <https://github.com/facebookresearch/detectron2> Face-



book Detectron2. Faster R-CNN / Mask R-CNN baseline framework; referenced in the project description.

- <https://universe.roboflow.com>  
Roboflow dataset hub. YOLO-format fire/smoke datasets for augmenting WWF annotations; also provides nighttime fire image collections.
- <https://github.com/AnshP99/firenet>  
FireNet open-source implementation. Lightweight edge baseline to reproduce and benchmark on target hardware.
- <https://arxiv.org/abs/2112.08598>  
FIgLib & SmokeyNet arXiv preprint. Full paper, dataset documentation, and SmokeyNet architecture details.

## Full Paper Links and Resources

### Research Paper Links

1. [P1] **Computer Vision Techniques for Forest Fire Perception** — Martinez-de Dios et al., 2008  
<https://www.sciencedirect.com/science/article/pii/S0262885607001096>
2. [P2] **A Review on Early Wildfire Detection from UAVs Using Deep Learning** — Bouguet-taya et al., 2022  
<https://www.sciencedirect.com/science/article/pii/S0165168421003467>
3. [P3] **Fire Detection Using Infrared Images for UAV-Based Forest Fire Surveillance** — IEEE, 2017  
<https://ieeexplore.ieee.org/document/7991306>
4. [P4] **FireNet: A Specialized Lightweight Fire & Smoke Detection Model** — Jadon et al., 2019  
<https://arxiv.org/abs/1905.11922>
5. [P5] **Forest Fire Smoke Detection Based on Deep Learning and UAV Images** — Kim & Muminov, 2023  
<https://pmc.ncbi.nlm.nih.gov/articles/PMC10304711/>
6. [P6] **A YOLO-Based Approach for Fire and Smoke Detection** — IJACSA, 2024  
[https://thesai.org/Downloads/Volume15No1/Paper\\_9-A\\_Yolo\\_based\\_Approach\\_for\\_Fire\\_and\\_Smoke\\_Detection.pdf](https://thesai.org/Downloads/Volume15No1/Paper_9-A_Yolo_based_Approach_for_Fire_and_Smoke_Detection.pdf)
7. [P7] **Wildfire Smoke Detection System: Model Architecture, Training, and Dataset** — arXiv, 2023  
<https://arxiv.org/abs/2311.10116>
8. [P8] **Forest Fire and Smoke Detection Using Learning Without Forgetting** — Fire Ecology, 2023  
<https://link.springer.com/article/10.1186/s42408-022-00165-0>
9. [P9] **Early Fire and Smoke Detection: A Comprehensive Review** — MDPI Applied Sciences, 2025  
<https://www.mdpi.com/2076-3417/15/18/10255>
10. [P10] **Benchmarking Multi-Scene Fire and Smoke Detection** — arXiv, 2024  
<https://arxiv.org/abs/2410.16631>
11. [P11] **FIgLib & SmokeyNet: Dataset and Model for Wildland Fire Smoke Detection** — Remote Sensing, 2022  
Dataset: <http://hpwren.ucsd.edu/HPWREN-FIgLib/>
12. [P12] **Fire and Smoke Datasets in 20 Years: An In-Depth Review** — arXiv, 2025  
<https://arxiv.org/abs/2503.14552>
13. [P13] **Real-Time Aerial Fire Detection Using Knowledge Distillation** — arXiv, 2025  
<https://arxiv.org/abs/2502.20979>
14. [P14] **Uncertainty-Aware Post-Detection Framework for Fire and Smoke Detection** —

arXiv, 2025

<https://arxiv.org/abs/2510.10108>

15. **[P15] Enhancing Forest Fire Detection Using Zero-Shot Learning: CLIP and LLaVA** — Springer, 2026  
<https://link.springer.com/article/10.1007/s11042-026-21377-5>

### GitHub Repositories

1. **Ultralytics YOLOv8/v11** — Primary detection backbone with TensorRT / ONNX export for Jetson Nano.  
<https://github.com/ultralytics/ultralytics>
2. **Ultralytics YOLOv5** — Compact nano/small variants for edge-baseline experiments.  
<https://github.com/ultralytics/yolov5>
3. **OpenAI CLIP** — ViT-B/32 visual-language embeddings for cloud/smoke disambiguation and nighttime glow classification.  
<https://github.com/openai/CLIP>
4. **Facebook Detectron2** — Faster R-CNN / Mask R-CNN framework; referenced baseline in the project description.  
<https://github.com/facebookresearch/detectron2>
5. **FireNet (AnshP99)** — Open-source FireNet with training dataset; lightweight edge baseline to reproduce.  
<https://github.com/AnshP99/firenet>
6. **Roboflow Universe** — YOLO-format fire/smoke datasets including nighttime collections for supplementing WWF data.  
<https://universe.roboflow.com/>